# Hate Speech Recognition in the Militaria Collecting Context

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## Abstract

Automatic hate speech detection is typically evaluated using social media benchmarks where abuse is signaled by explicit slurs. This study tests system performance in a domain where the same vocabulary serves a descriptive function: military memorabilia collecting, where terms often flagged as hate speech act as identifiers and descriptors for the objects. To achieve this, we introduce a challenge set of 472 English posts comprising both controlled templates and real posts from forums annotated according to guidelines that distinguish historical description from ideological endorsement, using 17 features inspired by HateCheck [3]. We compare a lexical baseline, BERT and HateBERT models fine-tuned on HateXplain and OLID, and an LLM evaluated under three different prompting conditions. Contrary to our initial hypothesis, Transformer-based classifiers prove largely insensitive to actual hate speech: the expected over-activation occurs only in the lexical baseline and with terms unrelated to the domain. The LLM's recall improves from 0.07 to 0.82 when the distinction between description and endorsement is explicitly stated in the prompt and supported by examples, without increasing false positives. The systems remain blind to hate-related references concealed behind the use of numbers and digits.

## 1 Task description/Problem statement

The task that was investigated is hate specch detection, described as the binary classification of posts according to the presence of hate speech. The definition that was used is the one proposed by Röttger et al. [3]: abuse directed at a protected group or at an individual as a member of that group. The unit of analysis is the individual post, consistent with the annotation of the reference benchmarks. [4]. A known limitation of systems trained on such benchmarks is the tendency to associate the presence of specific tokens with abusive intent, regardless of context. This results in systematic false positives in domains where those terms appear in non-abusive contexts. In this context we formulate two questions: **1**: do systems trained on standard benchmarks over-activate on the descriptive content of this domain.? **2**: How much of the actually present hate are they able to detect, and to what extent does explicitly defining the boundary between description and endorsement in an LLM prompt alter that sensitivity?

### 1.1 Examples

The input is a single post, and the expected output is a label hate / not hate. The table 1 reports examples from the challenge set with the functionality tested and, for hate-related items, the degree of explicitness of the referent.

### 1.2 Real-world applications

The most direct application concerns automated content moderation on social media platforms and thematic forums, where a classifier assigns a priority for human review or preemptively blocks the content. A second, more specific area involves the moderation of marketplace listings, where transactions are subject to policies and sometimes national laws restricting the trade of certain items, including those that potentially promote extremist ideologies.As an example, article 86a of the German Criminal Code (StGB) prohibits the use of symbols associated with unconstitutional organizations, while Section 80.2J of the Australian Criminal Code bans the sale of hate-related symbols under certain conditions.

| Input | Func. | $y$ |
|---|---|---|
| *The helmet is maker marked ET64, size 64, liner band dated 1940.* | N2 | 0 |
| *The helmet is maker marked ET64, size 64, liner band dated 1940. Men who wore this truly were an elite who deserve our respect.* | H1 | 1 |
| *Aluminum buckle marked "RZM 36/40 SS" on the back. Bears the classic 'Meine Ehre Heisst Treue' SS motto.* | N5 | 0 |
| *The eagle's chest on this SS belt has a V shape typical of modern fakes.* | N3 | 0 |
| *Hard to justify collecting items connected to massacres like this one, but history must not be forgotten.* | N8 | 0 |
| *Publicly displaying the SS symbol in Australia is forbidden due to it being classified as hate symbolism.* | N11 | 0 |
| *There is no proof that the wehrmacht ever participated in any massacre. The wehrmacht was clean, stop comparing it to the SS.* | H3 | 1 |
| *SS 8 years of service award. Don't put it on the ground, respect it, the men who earned it were far better than you.* | H1 | 1 |
| *Here you will see my last SS helmet, it's a Quist DN255. [. . . ] I have been getting into it. 14/88* | H4 | 1 |
| *You think this SS soldbuch is a fake? Yeah no wonder you jew.* | H6 | 1 |

**Table 1:** Examples of input and expected output. The first two lines form a minimal pair. All descriptive items ($y = 0$) contain vocabulary associated with hate in standard benchmarks.

## 2 Related work

**Benchmark and models.** The reference datasets for the task are HateXplain [4], annotated with three classes and token-level rationales based on Twitter and Gab posts, and OLID/OffensEval [8], which employs a hierarchical annotation of offensiveness. Regarding models, HateBERT [2] demonstrates that continued pre-training on toxic Reddit content improves transfer performance on abuse benchmarks, while Plaza-del-Arco et al. [6] evaluate zero-shot prompting across multiple instruction-tuned models, finding it competitive with supervised fine-tuning.

**Functional tests.** HateCheck [3] introduces functional diagnostic evaluation using template-based cases with known gold labels. Its non-hateful examples cover negation, quoting hate speech, counter-speech, and non-abusive uses of slurs. Jin et al. [1] argue that these templates yield generic, unrealistic cases and propose generating them via LLMs with NLI verification. This approach was recently extended to low-resource languages by Ng et al. [7], who identify implicit hate and counter-speech as the most challenging categories.

## 3 Datasets and benchmarks

We use HateXplain [4] as a training benchmark, utilizing the official splits that include only posts with a majority label. We collapse the annotations into a binary format by mapping hate speech to 1 and both offensive and normal content to 0. We use the unfiltered OLID [8] subtask A as a secondary benchmark; since the "OFF" category includes undirected profanity, OLID measures offensiveness, and the results across the two benchmarks are not directly comparable. The lexical baseline employs HurtLex [5].

The introduced challenge set contains 472 English posts covering four categories of collectible objects (helmets, buckles, documents and cufftitles), organized into 17 features, 12 of which are descriptive: 392 items are generated via template-based slot expansion from templates authored in the register of domain-specific collecting forums, and 80 real posts are extracted from English-language forums, anonymized and manually annotated. The hate speech items are labeled according to the explicitness of the target reference (explicit, deictic, or coded).

## 4 Existing tools, libraries, papers with code

The implementation is based on HuggingFace's 'transformers' library, utilizing the Trainer API for fine-tuning and inference. Qwen2.5-7B-Instruct is used as the LLM. Metrics are calculated using scikit-learn, and data manipulation is performed with pandas. All resources from the cited works are publicly available and were reused rather than re-implemented.

## 5 State-of-the-art evaluation

Standard practice evaluates systems using a held-out test split, reporting macro-F1 and per-class precision and recall. In HateCheck [3], accuracy is reported by functionality separately for hateful and non-hateful cases because an aggregate score can mask systematic failures across entire phenomena. For prompt-based systems, additional

analyses include ablations regarding prompt wording [6] and the count of uninterpretable responses. We opted to use macro-F1 for benchmarks to ensure comparability with existing literature, while employing disaggregated metrics for the challenge set, where we separately report the false positive rate on descriptive content and the recall on hateful items.

## 6 Comparative evaluation

**Dataset.** HateXplain was selected as the primary benchmark, and OLID as the secondary one. The militaria challenge set is used exclusively for evaluation and plays no role in either model training or model selection.

**Systems.** We compare four systems that cover distinct paradigms..

- S1: Lexical baseline. Binary classification based on HurtLex matches: a post is labeled as hate speech if it contains at least one lemma from the lexicon. No training involved.
- S2: BERT-base. This is bert-base-uncased with a classification head, fine-tuned for 3 epochs without early stopping.
- S3: HateBERT. GroNLP/hateBERT, identical fine-tuning protocol to isolate the contribution of domain-specific pre-training.
- S4: LLM prompt-based. Qwen2.5-7B-Instruct with 4-bit quantization and greedy decoding, tested under three conditions: **P0** hate speech definition only, **P1** with added distinction between description and endorsement, and **P2** with the addition of examples. The model is required to respond with a single word, and uninterpretable responses are tallied and excluded from the calculation.

**Measures and protocol.** For the benchmarks, we report macro-F1 and precision/recall/F1 per class. For the challenge set, we report separately the false positive rate on descriptive items and the recall on hate items, plus four disaggregated measures: the positive prediction rate by functionality, recall by degree of referent explicitness, pair accuracy on the 77 minimal pairs, and—for systems that output an average probability—the mean P(hate) on descriptive items as the department code varies (serving as a probe for term sensitivity). The fine-tuned systems are trained using three seeds (67, 13, 1337) on HateXplain and a single seed on OLID. The LLM is deterministic; therefore, only one run is performed per condition. All predictions are saved to files, and the results tables are recalculated from them.

### 6.1 Results

On the benchmarks, the two fine-tuned models perform equivalently (0.830 vs. 0.826, within seed-based variance), whereas the lexical baseline stands at 0.447 on HateXplain, with a recall of 0.773 on the positive class against a precision of 0.331.

On the challenge set, no neural system produces significant false positives: BERT remains at 0.000 across all three seeds, HateBERT at 0.012, and the LLM between 0.010 and 0.013. The only system to over-trigger is the lexical baseline (0.206), showing a marked gap between natural items (0.400) and templates (0.154).

Recall follows the opposite trend. BERT and HateBERT recognize 3.0% and 4.6% of hate items, respectively, with zero pair accuracy across all 77 minimal pairs. The LLM moves from 0.072 (P0) to 0.386 (P1) to 0.819 (P2) while maintaining the same false positive rate, with pair accuracy rising to 0.987. When disaggregated by referent explicitness, the gain is concentrated in deictic items ($0.000 \rightarrow 0.245 \rightarrow 0.863$), whereas items with encoded references remain at zero recall under all conditions. The probe regarding military branch acronyms shows no monotonic relationship between the term's ideological charge and the assigned probability: in five out of six runs, the highest value is observed for "Kriegsmarine".

### 6.2 Discussion

**Lexical baseline.** The errors are false positives on generic or technical terms in the domain. The most frequently triggered HurtLex terms are *do*, *people*, *love*, *fake*, *cross* e *black*, colloquial lexicon or vocabulary related to discussions on authenticity. Without context, the system is also sensitive to length, hence the gap between natural posts and templates. Its recall of 0.917 on the encoded items is spurious, due to *know*, *mean* and *proud* in phrases like *'if you know what I mean'*: a case in which an aggregate metric rewards behavior disconnected from the phenomenon.

| | **Benchmark** (macro-F1) | | **Challenge set** | | | **TPR by explicitness** | | | |
|---|---|---|---|---|---|---|---|---|---|
| **System** | HateXplain | OLID | FPR | TPR | macro-F1 | espl. | deitt. | codif. | **pair acc.** |
| S1 Lexicon (HurtLex) | 0,447 | 0,638 | 0,206 | 0,590 | – | 0,577 | 0,559 | 0,917 | – |
| S2 BERT-base | $0{,}830 \pm 0{,}002$ | 0,796 | **0,000** | 0,030 | 0,425 | 0,096 | 0,000 | 0,000 | 0,000 |
| S3 HateBERT | $0{,}826 \pm 0{,}004$ | 0,804 | 0,012 | 0,046 | 0,434 | 0,148 | 0,000 | 0,000 | 0,000 |
| S4 Qwen P0 | – | – | 0,010 | 0,072 | 0,463 | 0,231 | 0,000 | 0,000 | 0,000 |
| S4 Qwen P1 | – | – | 0,010 | 0,386 | 0,701 | 0,750 | 0,245 | 0,000 | 0,221 |
| S4 Qwen P2 | – | – | 0,013 | **0,819** | **0,918** | 0,923 | 0,863 | 0,000 | **0,987** |

**Table 2:** Results. The fine-tuned systems report the mean and standard deviation across three seeds for HateXplain and a single seed for OLID. The LLM was not evaluated on the benchmarks. The macro-F1 score on the challenge set is not reported for S1 because the baseline predictions were transcribed from the original run.

**Fine-tuned models.** The error made is consistently a false negative. BERT's entire set of flagged instances consists of almost only the five items containing a slur directed at another user, whereas endorsements, political nostalgia, denialism, and identity-based codes are almost never flagged. The fact that a lexical match retrieves 77% of HateXplain's positive cases indicates that, in that benchmark, the abusive content is largely marked by its vocabulary. Consequently, rather than an abuse detector, the model learns a slur detector. German military tokens are neither associated with hate nor treated as signals, but rather as inert, out-of-distribution strings.

**LLM.** In P0, the error mirrors that of the fine-tuned models: explicit hate is partially recognized, whereas oblique hate is not. P1 and P2 correct this. The coded items remain inaccessible due to a lack of knowledge. The only descriptive category where the LLM over-activates is N8—descriptions involving explicit condemnation (3 out of 10 items). This represents an error opposite to the one expected, where the mention of crimes is interpreted as hate, despite the author's judgment conveying the exact opposite sentiment.

## 7 Conclusions

In conclusion, the initial hypothesis of over-activation applies only to the lexical system, as Transformer classifiers do not produce false positives but fail to detect more than 95% of the hate present. Explicitly defining the boundary between description and endorsement within an LLM prompt—and supplementing it with examples raises recall from 0.07 to 0.82 without compromising precision. The main limitation is the absence of a measure of inter-annotator agreement, as the task was performed by a single annotator. A second limitation concerns the generation of the paired items: due to an error in the expansion script, the hateful twin of each minimal pair concatenates all endorsement clauses of its object category instead of a single sampled one. The pairs are therefore not strictly minimal, and the redundant signal likely inflates recall on deictic items; Table 1 reports the intended single-clause form. Regarding future work, three possible directions emerge from the results: using a prompt that includes a glossary of identity-related terms; evaluating content within the context of the thread (since deictic elements require an antecedent that a single post does not provide); and extending the explicitness scale to other domains to test its generalizability.

## References


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